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Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MECHATRONIC SYSTEMS
NQF LEVEL 2

(6040102)

28 November 2025 (Y-paper)
13:00–16:00

This question paper consists of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA
NATIONAL CERTIFICATE (VOCATIONAL)
MECHATRONIC SYSTEMS
NQF LEVEL 2
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
 2. Read all the questions carefully.
 3. Number the answers according to the numbering system used in this question paper.
 4. Use only a black or blue pen.
 5. Write neatly and legibly.
-

2)

-3-

QUESTION 1

- 1.1 Electrical energy may be sourced and converted from various other energy forms.

List FIVE energy forms that may be converted to electrical energy. (5 × 1) (5)

- 1.2 FIGURE 1 below shows a diagram of a basic technical system.

Fill in the missing labels by writing only the answer next to the question number (1.2.1–1.2.4) in the ANSWER BOOK. (4 × 1) (4)

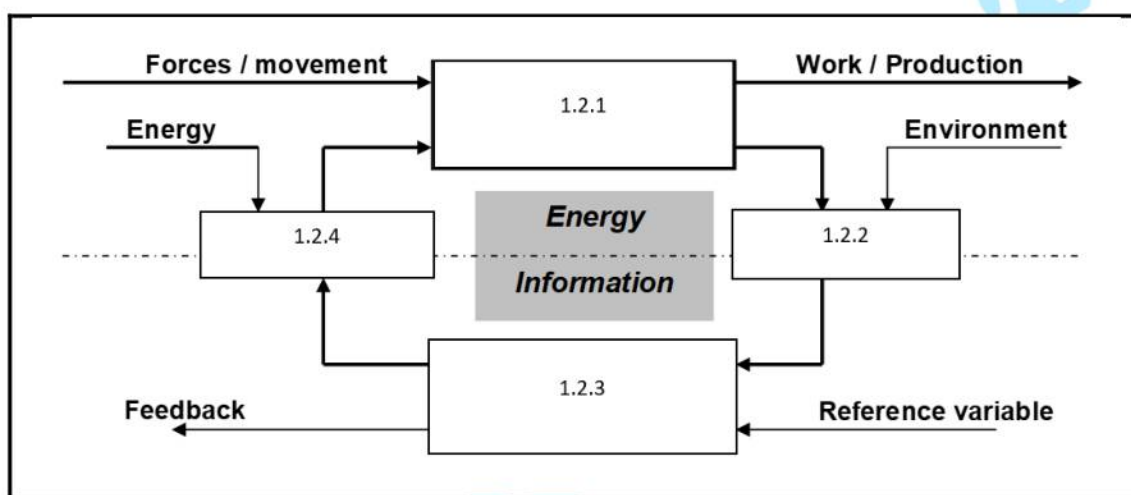


FIGURE 1

- 1.3 Show, by means of a drawing, the difference between a *hot* and *cold bimetal strip*. (2 × 2) (4)
- 1.4 Explain the importance of the carbon brushes in a DC motor. (4 × 1) (4)
- 1.5 Briefly explain the function of a worm and wheel gear transmission. (3 × 1) (3)

[20]

2)

-4-

QUESTION 2

- 2.1 Define the following terms
- | | | | |
|-------|-------------|---------|-----|
| 2.1.1 | Comparator | (2 × 1) | (2) |
| 2.1.2 | Truth table | (2 × 1) | (2) |
| 2.1.3 | Switch | (2 × 1) | (2) |
- 2.2 Draw a labelled symbol of a NOT gate using S1 as an input and E1 as an output.
(3 × 1) (3)
- 2.3 Draw the symbol for a magnetic reed switch.
(4 × 1) (4)
- 2.4 Name the FIVE steps to follow when implementing a problem-solving technique.
(5 × 1) (5)
- 2.5 Write the following acronyms in full.
- | | | |
|-------|-----|-----|
| 2.5.1 | CNC | (1) |
| 2.5.2 | CIM | (1) |
- [20]**

QUESTION 3

- 3.1 Define the term, *compressed air*.
(2 × 1) (2)
- 3.2 List FOUR characteristics of compressed air.
(4 × 1) (4)
- 3.3 Draw a non-simplified service unit.
(7 × 1) (7)
- 3.4 Draw a double-acting cylinder indicating the unequal forces when it goes positive and negative.
(3 × 1) (3)
- 3.5 What is a two-pressure valve mainly used for?
(4 × 1) (4)
- [20]**

2)

-5-

QUESTION 4

- 4.1 Mention FOUR safety devices that are used for over-current protection. (4)
- 4.2 FIGURE 2 below shows a circuit diagram of a control box. Study the circuit diagram and answer the questions that follow.

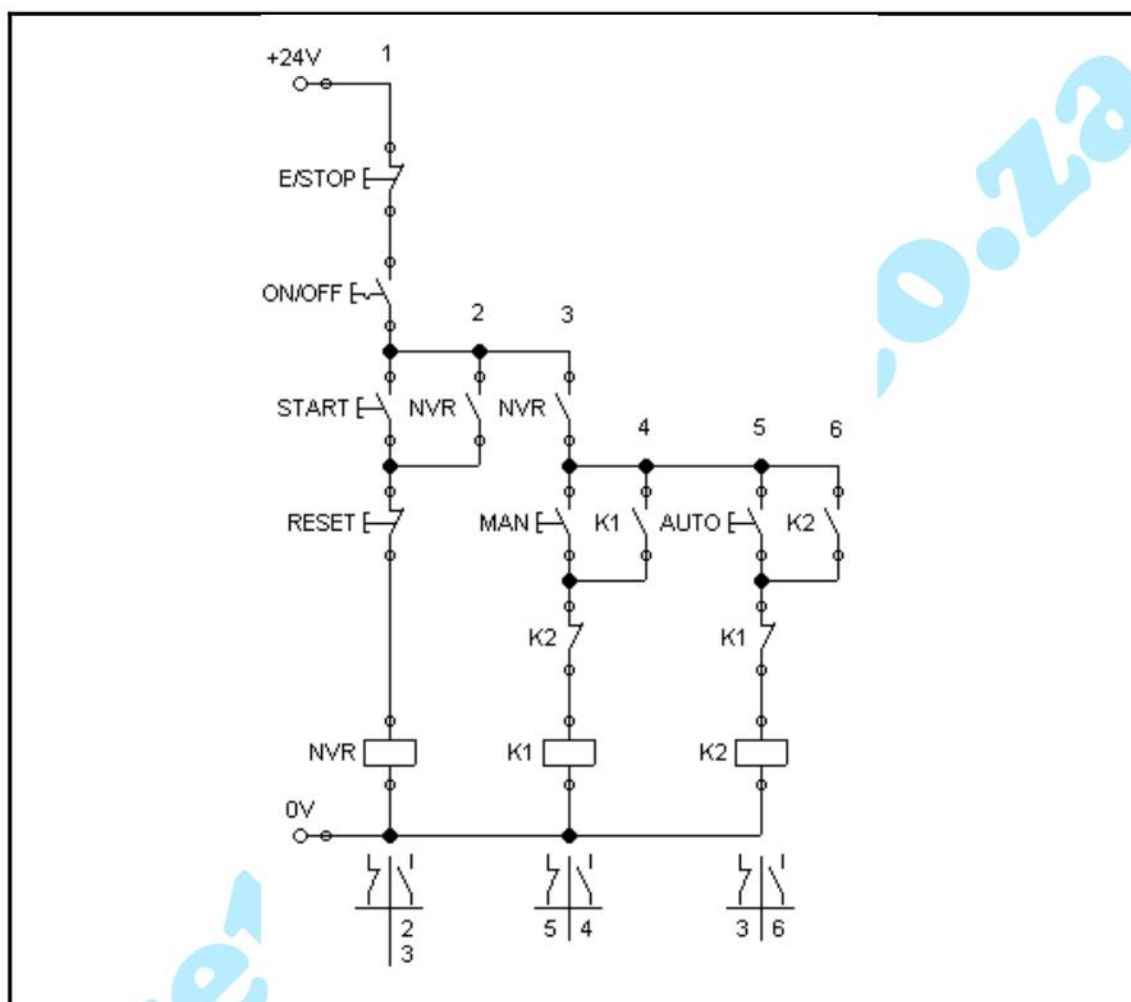


FIGURE 2

- 4.2.1 Explain the purpose of the NVR in this circuit. (2 × 1) (2)
- 4.2.2 Explain why an E/Stop is used in the circuit. (2 × 1) (2)
- 4.2.3 Identify the TWO contacts that prevent the manual and automatic controls from being activated at the same time. (2 × 1) (2)
- 4.2.4 Explain the operation of the circuit shown in the diagram. (7 × 1) (7)
- 4.3 Safety voltage results in low risk of shock.
Give the voltage that is used in electro-pneumatic control and state how this voltage is achieved from the supply voltage. (3 × 1) (3)

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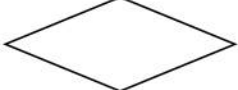
-6-

QUESTION 5

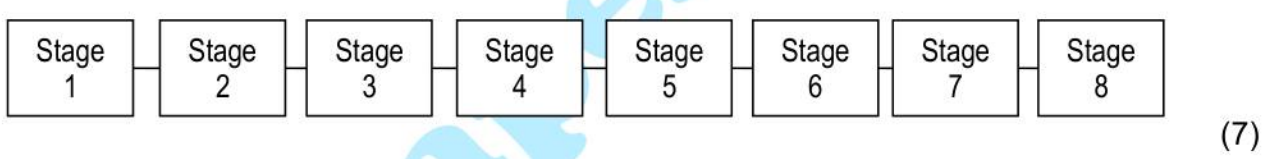
- | | | | |
|-----|---|---------|-----|
| 5.1 | List FOUR faults that may occur on a service unit. | (4 × 1) | (4) |
| 5.2 | Mention THREE faults that may occur on tubing and pipes. | (3 × 1) | (3) |
| 5.3 | Briefly explain why maintenance is important. | (2 × 1) | (2) |
| 5.4 | What are the following features used for in a flow chart? | | |

5.4.1		(1 × 1)	(1)
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5.4.2		(1 × 1)	(1)
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5.4.3		(1 × 1)	(1)
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- | | | | |
|-----|---|--|--|
| 5.5 | Copy and complete the diagram below in the ANSWER BOOK to show how one would do half-cut testing. | | |
|-----|---|--|--|



- | | | | |
|-----|---------------------------------|---------|-----|
| 5.6 | Write the acronym, SFC, in full | (1 × 1) | (1) |
|-----|---------------------------------|---------|-----|
- [20]**

TOTAL: 100